



Modelling Financial Markets

Chapter 14: Deep Learning and Time-Series Foundation Models



Daniel Traian PELE

Bucharest University of Economic Studies

IDA Institute Digital Assets

Blockchain Research Center

AI4EFin Artificial Intelligence for Energy Finance

Romanian Academy, Institute for Economic Forecasting

MSCA Digital Finance

Acronyms used in this material (1/2)

- ▣ ACF: AutoCorrelation Function
- ▣ ACI: Adaptive Conformal Inference
- ▣ AI: Artificial Intelligence
- ▣ AR: AutoRegressive (model)
- ▣ ARFIMA: AutoRegressive Fractionally Integrated Moving Average (model)
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ CI: Confidence Interval
- ▣ COVID-19: COronaVirus Disease 2019
- ▣ CPU: Central Processing Unit
- ▣ CRPS: Continuous Ranked Probability Score
- ▣ CRSP: Center for Research in Security Prices
- ▣ DM: Diebold–Mariano (test)
- ▣ DQ: Dynamic Quantile (test)
- ▣ ES: Expected Shortfall
- ▣ ETF: Exchange-Traded Fund
- ▣ FHS: Filtered Historical Simulation
- ▣ FRED: Federal Reserve Economic Data
- ▣ FZ0: Fissler–Ziegel loss, zero-homogeneous version
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ GIFT: General Time Series Forecasting Model Evaluation (benchmark GIFT-Eval)
- ▣ GPD: Generalised Pareto Distribution
- ▣ GRU: Gated Recurrent Unit
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ HAR: Heterogeneous AutoRegressive (model)
- ▣ HS: Historical Simulation
- ▣ LASSO: Least Absolute Shrinkage and Selection Operator

Acronyms used in this material (2/2)

- LLM: Large Language Model
- LR: Likelihood Ratio (test)
- LSTM: Long Short-Term Memory (network)
- MCS: Model Confidence Set
- MDE: Minimum Detectable Effect
- MLP: MultiLayer Perceptron
- MSE: Mean Squared Error
- NBER: National Bureau of Economic Research
- PIT: Probability Integral Transform
- QLIKE: Quasi-LIKElihood (loss function)
- QS: Quantile Score
- RFF: Random Fourier Features
- RNN: Recurrent Neural Network
- RV: Realised Variance
- SPA: Superior Predictive Ability (test)
- TS: Time Series (in the name TS-Arena)
- US: United States
- VAR: Vector AutoRegression

Outline

Why Deep Learning for Financial Time Series?

Neural Networks for Sequences

Attention and Transformers

Time-Series Foundation Models

Test 1: Forecasting Returns

Test 2: Forecasting Realised Volatility

Test 3: VaR 1% and ES 2.5%

Case Study: The Virtue of Complexity

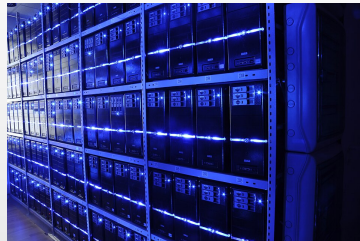
AI for Scientific Discovery

Practice and Conclusions

References

The Promise and the Question

- **Deep learning:** models built from many layers of simple non-linear units, trained on data
 - ▶ It transformed image recognition and language processing
 - ▶ Since 2024: **foundation models** for time series, pretrained on billions of observations
- The question of this chapter
 - ▶ Do these models forecast **returns**, **volatility** and **tail risk** better than the classical models of Chapters 5–9?
 - ▶ We answer with the evaluation tools of Chapter 8



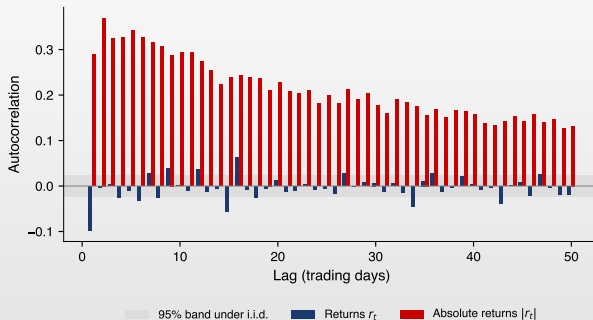
A data centre: deep networks are trained on racks of such servers

Photo: [BalticServers.com](#) (2013); CC BY-SA 3.0; [Wikimedia Commons](#)

Three Forecasting Targets

- ▣ **Returns** r_t (conditional mean)
 - ▶ Very low signal-to-noise ratio; predictable patterns are traded away (Chapter 2)
- ▣ **Volatility** (conditional variance)
 - ▶ Persistent and clustered (Chapters 1 and 5): there is a signal to learn
- ▣ **Tail-risk measures**: VaR 1% and ES 2.5%
 - ▶ Need the shape of the lower tail, not just the centre of the distribution (Chapter 7)
- ▣ Each target gets its own loss function and its own benchmark model

Where Is the Signal?



- S&P 500 daily returns since 2000 (6 718 days): autocorrelation at lags 1 to 50
- Grey band: 95% range under independence, $\pm 1.96/\sqrt{T} = \pm 0.024$

Interpretation: Mean versus Size of Moves

- ▣ Returns: first-order autocorrelation -0.099
 - ▶ 15 of 50 lags outside the band, but small and of changing sign: little to learn about the mean
- ▣ Absolute returns: autocorrelation 0.29 at lag 1, still 0.13 at lag 50
 - ▶ All 50 lags outside the band: volatility clustering with long memory
- ▣ Prediction for the rest of the chapter: any flexible model can find the volatility signal; none should find much in the mean

What We Do in This Chapter

- ▣ Neural networks for sequences: vanishing gradients as VAR impulse responses, the LSTM as an AR(1)
- ▣ Attention and Transformers: patches, positions, quantile outputs
- ▣ Foundation models: Chronos, TimesFM, Moirai; zero-shot use and pretraining leakage
- ▣ An evaluation protocol: contamination, nested and multiple comparisons, calibration, power
- ▣ Three empirical tests: returns, realised volatility, VaR 1% and ES 2.5%

Section Summary

- ▣ Three targets with very different signal-to-noise ratios: mean, volatility, tail
- ▣ Absolute returns are strongly autocorrelated, returns are not
- ▣ Every new model is judged against a simple benchmark with the tests of Chapter 8

Prerequisites Recap: MLP, Backpropagation, Adam

- Multilayer perceptron (MLP): $h^{(\ell)} = \varphi(W^{(\ell)}h^{(\ell-1)} + b^{(\ell)})$, $h^{(0)} = x$; activations ReLU $\max(0, z)$, tanh, sigmoid $\sigma(z) = 1/(1 + e^{-z})$
 - ▶ Trained by backpropagation (Rumelhart, Hinton & Williams (1986)) and Adam (Kingma & Ba (2015)) on mini-batches; loss MSE for points, pinball $(u - 1\{y < q\})(y - q)$ for the quantile q of level u
- **Universal approximation** (Hornik, Stinchcombe & White (1989)): one hidden layer approximates any continuous function
 - ▶ Silent on the sample size needed: the binding constraint with one financial series
- Regularisation: early stopping, weight decay $\lambda\|\theta\|^2$, dropout, ensembles over random seeds
 - ▶ Finance-specific: the validation block must follow the training block (Chapter 13, purged cross-validation)
 - ▶ Our LSTM has 1 233 parameters; Chronos-2 about 119 million

Deep Learning Comes of Age

- ▣ [LeCun, Bengio & Hinton \(2015\)](#): review of the methods behind the breakthroughs in vision and speech
- ▣ 2024: Nobel Prize in Physics to Hopfield and Hinton for foundational work on neural networks
- ▣ In finance: [Gu, Kelly & Xiu \(2020\)](#) find that neural networks improve monthly stock-return predictions through non-linear interactions (Chapter 13)
 - ▶ Cross-section of thousands of stocks, not one index: much more data

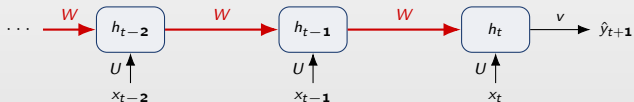


Geoffrey Hinton, Nobel laureate 2024

Photo: [Arthur Petron \(2024\)](#); CC BY-SA 4.0; [Wikimedia Commons](#)

RNN and LSTM in One Slide

- ▣ **Recurrent neural network (RNN):** $h_t = \tanh(W h_{t-1} + U x_t + b)$, $\hat{y}_{t+1} = v^\top h_t + c$
 - ▶ The same weights at every step: unrolled over time, a very deep network with shared weights
- ▣ **Long short-term memory (LSTM, Hochreiter & Schmidhuber (1997)):** gates $f_t, i_t, o_t = \sigma(W_\bullet x_t + U_\bullet h_{t-1} + b_\bullet)$, candidate $g_t = \tanh(W_g x_t + U_g h_{t-1} + b_g)$
 - ▶ Cell state $c_t = f_t \odot c_{t-1} + i_t \odot g_t$, output $h_t = o_t \odot \tanh(c_t)$; $\odot$: element-by-element product
 - ▶ f_t : forget gate, i_t : input gate, o_t : output gate; GRU (gated recurrent unit): a lighter variant with two gates



Vanishing Gradients: a Bound, Not an Anecdote

- Backpropagation through time multiplies one Jacobian per step

- ▶ $\frac{\partial h_T}{\partial h_{T-k}} = \prod_{j=0}^{k-1} D_{T-j} W, \quad D_t = \text{diag}(\varphi'(Wh_{t-1} + Ux_t + b))$

- Submultiplicativity of the spectral norm gives the bound

- ▶ $\|\partial h_T / \partial h_{T-k}\|_2 \leq (\gamma_\varphi \|W\|_2)^k, \quad \gamma_\varphi = \sup_z |\varphi'(z)| \quad (= 1 \text{ for } \tanh)$
 - ▶ $\gamma_\varphi \|W\|_2 < 1$ suffices for geometric vanishing, so $\gamma_\varphi \|W\|_2 \geq 1$ is necessary for explosion (Pascanu, Mikolov & Bengio (2013))
 - ▶ Spectral radius ρ : $\rho(W) > 1$ is necessary only for a linear RNN; with φ the local Jacobians $D_t W$ matter ($\rho(W) = 0$ yet $\rho(DW) > 1$ is possible)
- Scalar case $|w|^k$: $0.9^{20} = 0.12, 0.9^{50} = 0.005$ (Bengio, Simard & Frasconi (1994))
 - ▶ Remedies: clipping the gradient norm (explosion); gating, as in the LSTM (vanishing)

A Linearised RNN Is a VAR(1)

- Linearise around a fixed point h^* with $D = \text{diag}(\varphi')$ evaluated there
 - ▶ $h_t - h^* \approx A(h_{t-1} - h^*) + DUx_t$, $A = DW$
 - ▶ A VAR(1) (vector autoregression) in the hidden state, with innovation DUx_t
- The Jacobian $\partial h_T / \partial h_{T-k} = A^k$ is the impulse response of this VAR at horizon k
 - ▶ Stable VAR, $\rho(A) < 1$: impulse responses and gradients both decay like $\rho(A)^k$
 - ▶ Explosive VAR, $\rho(A) > 1$: gradients explode
- Consequence: a stable RNN is a short-memory model by construction, like any stationary VAR

Why the LSTM Remembers

- Along the cell state the only multiplier is the forget gate
 - ▶ $\frac{\partial c_T}{\partial c_{T-k}} = \prod_{j=0}^{k-1} f_{T-j}$ (direct path)
 - ▶ $c_t = f_t \odot c_{t-1} + i_t \odot g_t$: a random-coefficient AR(1) in c_t
- Constant forget gate $f = \sigma(b_f)$: an AR(1) with coefficient f
 - ▶ $b_f = 0$: $f = 0.5$, $0.5^{50} \approx 9 \times 10^{-16}$
 - ▶ $b_f = 3$: $f = 0.953$, $0.953^{50} \approx 0.09$; half-life 14 steps



Sepp Hochreiter, co-inventor of the LSTM

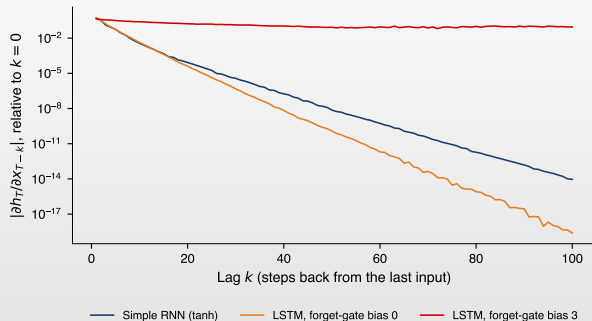
Photo: Eulenreich (2015); CC BY-SA 4.0; Wikimedia

[Commons](#)

Short Memory by Construction: Geometric versus Hyperbolic Decay

- Constant gates, i.i.d. candidate input g_t , stationary start: $\text{Corr}(c_t, c_{t-k}) = f^k$, geometric decay
 - ▶ Long memory (ARFIMA with $0 < d < 1/2$) decays hyperbolically: $\rho_k \sim Ck^{2d-1}$
- S&P 500 absolute returns since 2000: ACF 0.289 at lag 1, 0.132 at lag 50
 - ▶ Geometric decay with its own amplitude, $\rho_k = \rho_1 f^{k-1}$ (an AR(1) signal observed with noise), matches both lags with $f = 0.984$: $b_f = 4.1$, half-life 43 days
 - ▶ That f implies 0.027 at lag 150; the data show 0.059, more than twice as much (95% band ± 0.024)
- A constant gate $f < 1$ is a short-memory filter; a long context or explicit long averages, as in HAR (Test 2), only approximate long memory over a finite range of lags

The Vanishing Gradient, Measured



- ▣ $|\partial h_T / \partial x_{T-k}|$ by automatic differentiation; 16 hidden units, 50 random initialisations, log scale
- ▣ The same random inputs for the three networks; only the architecture and the forget-gate bias change

Interpretation: Architecture Is Not Enough

- ▣ Simple RNN: gradient share 8.3×10^{-5} at lag 20 and 7.4×10^{-9} at lag 50
 - ▶ Practically nothing is learned from inputs more than a few weeks back
- ▣ LSTM with forget-gate bias 0: 1.2×10^{-10} at lag 50, no better than the RNN
 - ▶ At initialisation $f \approx 0.5$: the gate halves the memory at every step
- ▣ LSTM with forget-gate bias 3: 8.4% at lag 50, 9.0% at lag 100
 - ▶ Matches the constant-gate calculation $0.953^{50} \approx 0.09$
 - ▶ Practical rule: initialise the forget-gate bias at a positive value

How Big Is a Small LSTM?

- Parameters of one LSTM layer with input size d and hidden size h
 - ▶ $4 \times (hd + h^2 + h)$ (PyTorch uses two bias vectors: $4 \times (hd + h^2 + 2h)$)
 - ▶ Four blocks: three gates and the candidate
- Our return model: $d = 1$, $h = 16$, linear output: 1 233 parameters
 - ▶ Trained on at most 4,000 daily returns: about three observations per parameter
- Application in finance: [Fischer & Krauss \(2018\)](#) use LSTMs to predict whether S&P 500 constituents beat the cross-sectional median the next day

Section Summary

- ▣ A linearised RNN is a VAR(1): gradients vanish at the rate at which its impulse responses decay
- ▣ The LSTM cell state is an AR(1) with coefficient f_t : memory lasts only as long as the forget gate allows, and decays geometrically
- ▣ Even small LSTMs have thousands of parameters: a lot for one financial series

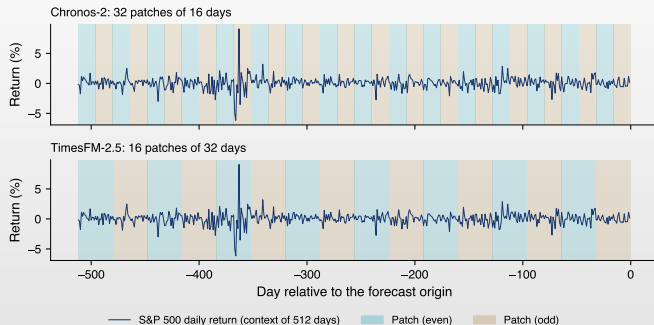
Attention: Looking Back Directly

- Each time step asks a question (query) and compares it with every past step (key)
 - ▶ $\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V$ (Vaswani et al. (2017))
 - ▶ $Q = XW_Q$, $K = XW_K$, $V = XW_V$: linear projections of the inputs X ; d_k : key dimension
 - ▶ softmax turns scores into weights that sum to one; division by $\sqrt{d_k}$ keeps the scores from saturating it
- No product of Jacobians over time: any past step is one hop away
 - ▶ Cost grows with the square of the number of inputs n : $O(n^2)$

Positions and Patches

- **Positional encoding:** attention treats its inputs as a set, so the order must be added
 - ▶ Sinusoids of different frequencies added to the inputs (original Transformer)
 - ▶ Rotary embeddings: rotate queries and keys by an angle proportional to the position (used by Chronos-2)
- **Patching** (Nie et al. (2023)): a token is a block of consecutive observations, not a single one
 - ▶ 512 daily returns in patches of 16 days: 32 tokens instead of 512, about 256 times fewer attention scores
 - ▶ Each token summarises local dynamics (a few weeks of market activity)

Patching a 512-Day Context



- The last 512 daily returns of the S&P 500, split as Chronos-2 (16 days) and TimesFM-2.5 (32 days) read them
- Each shaded block becomes one embedding vector for the Transformer

Section Summary

- ▣ Attention compares every step with every other step: no vanishing gradient through time, but quadratic cost
- ▣ Positions must be encoded; patches reduce the number of tokens and carry local context
- ▣ DeepAR ([Salinas et al. \(2020\)](#)), N-BEATS ([Oreshkin et al. \(2020\)](#)), PatchTST ([Nie et al. \(2023\)](#)): learn across many series; foundation models push this to its limit

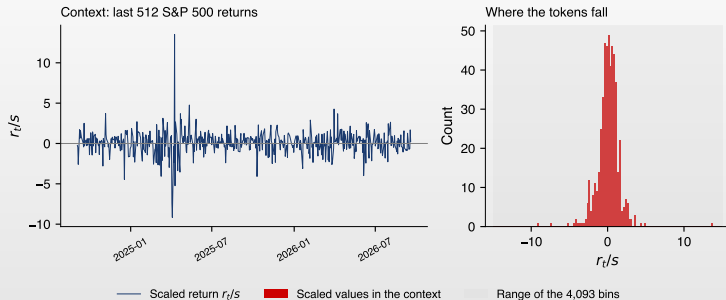
What Is a Foundation Model?

- ▣ A large network pretrained once on a very broad corpus, then used on new tasks
 - ▶ Idea borrowed from large language models (LLM): pretrain on text, then use on any question
- ▣ **Zero-shot** forecasting: no training on the target series; only its recent history (the **context**) is passed at forecast time
 - ▶ Output: a set of quantiles of the future values, not only a point
 - ▶ Fine-tuning (further training on the target series) is possible but not used here
- ▣ Appeal for finance: one model for thousands of series, no estimation per series

Chronos: Time Series as a Language

- ▣ [Ansari et al. \(2024\)](#): turn values into tokens and train a language model (T5) with cross-entropy
 - ▶ Mean scaling: divide the context by $s = \frac{1}{C} \sum_t |x_t|$
 - ▶ Quantisation: 4,093 equally spaced bin centres in $[-15, 15]$ (width 0.0073), plus special tokens
 - ▶ Forecast: sample future tokens, map back to values, read quantiles from the samples
- ▣ Pretraining corpus: public datasets (energy, transport, retail, weather, web, finance and economics) plus synthetic series (TSMixup, KernelSynth)
 - ▶ Model sizes 20 to 710 million parameters; the small version has 46 million

What Chronos Sees: Scaled Returns and Tokens



- Context: the last 512 S&P 500 daily returns; scale $s = 0.672\%$ (mean absolute return)
- Right: where the scaled values fall within the token range $[-15, 15]$

Interpretation: a Vocabulary Mostly Unused

- ▣ Scaled returns range from -9.2 to 13.5; 96% lie within ± 3
 - ▶ The 512 returns use 336 distinct tokens out of 4,093; ± 3 alone spans 818 bins
- ▣ Scaling makes the model free of units: 1% in a calm year and 3% in a crisis can map to the same token
 - ▶ Good for transfer across series; the level of volatility must be recovered from the context itself
- ▣ A crash day M raises s by about $|M|/C$ and compresses all other days: strongly in a short context, little in a 512-day one (Seminar, A3–A4)

Chronos-Bolt and Chronos-2

- ▣ **Chronos-Bolt:** patches of 16 values and a direct quantile output instead of token sampling
 - ▶ Quantile levels 10%, 20%, ..., 90% only; small version 48 million parameters
- ▣ **Chronos-2** ([Ansari et al. \(2025\)](#)): robust scaling, patches of 16, attention across time and across groups of related series
 - ▶ 21 quantile levels: 1%, 5%, 10%, ..., 95%, 99%; about 119 million parameters
 - ▶ Training corpus: selected datasets of the Chronos and [GIFT-Eval](#) corpora plus synthetic multivariate series
 - ▶ Benchmarks: [GIFT-Eval](#) ([Aksu et al. \(2024\)](#)) and fev-bench

TimesFM, Moirai and Lag-Llama

- **TimesFM** (Das et al. (2024)): decoder-only Transformer on patches of 32 values
 - ▶ Pretrained on about 10^{11} time points: Google Trends, Wikipedia page views, synthetic series, M4, electricity, traffic, weather
 - ▶ TimesFM-2.5: mean plus deciles; the released checkpoint has about 231 million parameters
- **Moirai** (Woo et al. (2024)): any number of variables, mixture-distribution output; Moirai 2.0 (Liu et al. (2025))
 - ▶ Released under a non-commercial licence; not used in our tests
- **Lag-Llama** (Rasul et al. (2023)): decoder-only model with lagged values as features and a Student- t output

The Three Models We Test

Model	Parameters	Patch	Quantile levels	Weights published
Chronos-Bolt (small)	48 M	16	10%–90%	25.11.2024
TimesFM-2.5	231 M	32	mean, 10%–90%	01.10.2025
Chronos-2	119 M	16	1%, 5%, . . . , 99%	30.10.2025

- All three run on an ordinary CPU; context of 512 daily observations, one-day-ahead forecasts
- Publication dates of the weights from the Hugging Face model repositories; they fix what the models could have seen
- Only Chronos-2 reaches the 1% level needed for VaR 1%

Pretraining Leakage

- A test is out of sample only if the test period was never seen during training
 - ▶ The Chronos corpus includes financial series (e.g. daily exchange rates 1990–2016); later corpora are broader
 - ▶ We cannot rule out that our series, or series driven by the same shocks, were in the pretraining data
- Our remedy: a **post-release window** from 3 November 2025, after all three sets of weights were published
 - ▶ Clean, but short (about 220 trading days; 320 calendar days for Bitcoin): low power, especially for VaR 1%
- Look-ahead in our own pipeline is avoided too: every scaling, window and estimate uses only past data

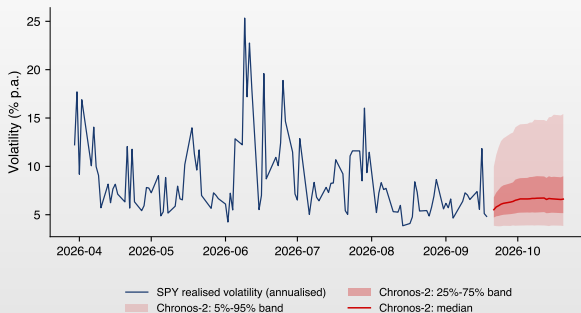
From a Quantile Grid to What We Need

- The model returns quantiles q_u at levels $u_1 < \dots < u_K$
 - ▶ Point forecast: the median $q_{0.5}$, or the mean $\int_0^1 q_u du$ approximated on the grid
 - ▶ Mean of a transformed variable: $E[g(Y)] = \int_0^1 g(q_u) du$ for increasing g (e.g. $g = \exp$ for log variance)
- Tail risk (Chapter 7): $\text{VaR}_\alpha = -q_\alpha$, $\text{ES}_\alpha = -\frac{1}{\alpha} \int_0^\alpha q_u du$
 - ▶ Below the lowest level the grid is silent: holding q_u at its last value understates the tail part of ES; interpolation between levels has no fixed sign
 - ▶ Chronos-Bolt and TimesFM stop at 10%: a "VaR 1%" request returns the 10% quantile

Worked Example: the Last Chronos-2 Forecast

- S&P 500, forecast for the day after 18.09.2026: $q_{0.01} = -2.70\%$, $q_{0.05} = -1.40\%$
 - ▶ VaR 1% = $-q_{0.01} = 2.70\%$
 - ▶ Linear interpolation: $q_{0.025} = q_{0.01} + \frac{0.015}{0.04}(q_{0.05} - q_{0.01}) = -2.21\%$, so VaR 2.5% = 2.21%
- ES 2.5%: $-\frac{1}{0.025} \left[0.01 q_{0.01} + 0.015 \frac{q_{0.01} + q_{0.025}}{2} \right] = 2.56\%$
 - ▶ A tail-truncated approximation: cutting the tail at $q_{0.01}$ lowers ES, the interpolated part can err either way (Seminar, A6)

A Zero-Shot Fan Chart for Volatility



- Chronos-2 on log realised variance of SPY (512-day context), 22 trading days ahead from 18.09.2026; annualised volatility
- Bands: quantiles 5%–95% and 25%–75% of the predictive distribution

Interpretation: Mean Reversion Learned Without Training

- Last observed volatility 4.8% p.a.; average of the last 20 days 6.6%
 - ▶ Median forecast at 22 days: 6.6%, with a 90% band from 3.8% to 15.4%
- The band widens with the horizon and is skewed upwards: volatility is bounded below by zero but can rise sharply
 - ▶ Features of GARCH and HAR forecasts (Chapters 5 and 9), obtained here with no estimation on SPY data
- Whether these forecasts are **better** is an empirical question: next sections

Contamination: a Taxonomy

- ▣ Four ways in which a test period can be “seen” before the forecast
 - ▶ (i) The same series is in the pretraining corpus over the test period
 - ▶ (ii) Correlated series (other indices, exchange rates) over the test period carry the same shocks
 - ▶ (iii) Developers select architectures and checkpoints on public benchmarks that overlap our test
 - ▶ (iv) Our own look-ahead: scalings, windows, seeds or contexts chosen with test data
- ▣ Only (iv) is fully under our control; a post-release window removes (i)–(iii) for the weights we use
 - ▶ Electricity prices: [Pan & Ezzat \(2026\)](#) benchmark on two datasets to limit contamination; foundation models are competitive but do not consistently beat domain-specific methods

A Formal Protocol: Pre/Post Test, Live Benchmarks, Ablations

- Pre/post regression on the loss differential $d_t = L_t^{FM} - L_t^{bench}$: $d_t = \beta_0 + \beta_1 \mathbf{1}\{t \geq \tau\} + u_t$, HAC s.e.
 - ▶ τ : release date of the weights (a common cutoff when several models are compared); memorisation predicts $\beta_1 > 0$; H_0 : $\beta_1 = 0$
 - ▶ Report the minimum detectable effect (MDE) of β_1 : with 218 post-release days it is 4.9 times the pre-release gap (last section)
- Pre-registration: forecasts submitted before the outcome exists make contamination impossible by design
 - ▶ Live benchmarks: TS-Arena ([Meyer et al. \(2025\)](#)), Impermanent ([Garza et al. \(2026\)](#)); our post-release window is a manual version
- Ablations: removing the language-model part of LLM-based forecasters does not degrade accuracy, often improves it ([Tan et al. \(2024\)](#))

What the Literature Finds on Realised Volatility

- [Goel et al. \(2025\)](#): TimesFM on volatility; zero-shot a reasonable baseline, incremental fine-tuning needed to beat econometric models (DM and Giacomini–White tests)
- [Brini \(2026\)](#): nine zero-shot foundation models against eight HAR-type models, 50 assets, three horizons
 - ▶ Pooled losses favour foundation models, but the gain comes from a few outlier assets
 - ▶ Averaging loss ratios to log-HAR asset by asset, only one small model beats log-HAR at every horizon, narrowly
- Our Test 2 is one asset of this design: expect parity with log-HAR, not dominance

Section Summary

- ▣ Foundation models: pretrained once on large corpora, used zero-shot with a context window
- ▣ They read a series through scaling, tokens or patches, and return quantiles
- ▣ Quantile grids limit tail estimates; contamination of types (i)–(iii) requires a post-release window or pre-registration

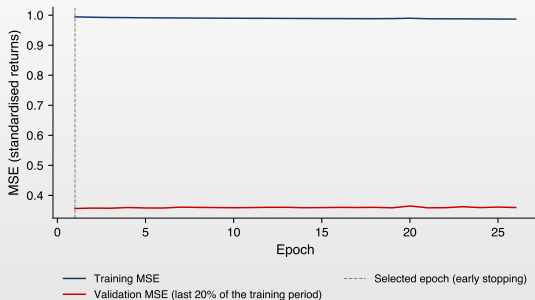
Design of the Return Test

- ▣ Daily log returns (%): S&P 500 and BET from 04.01.2016 (2 693 and 2 675 days), Bitcoin from 01.01.2019 (2 818 days)
 - ▶ Every forecast uses only data up to the previous close
- ▣ Models: historical mean, AR(1) on a 1,000-day window, LSTM, Chronos-2, Chronos-Bolt and TimesFM-2.5 (median)
 - ▶ Squared error targets the mean; grid means ($\int_0^1 q_u du$) do worse still, e.g. S&P 500 R^2 -0.77% (Chronos-2), -3.42% (TimesFM-2.5)
- ▣ Benchmark: the zero forecast; out-of-sample $R^2 = 1 - \sum_t (r_t - \hat{r}_t)^2 / \sum_t r_t^2$
 - ▶ Zero rather than the historical mean, as in [Gu, Kelly & Xiu \(2020\)](#); compare [Campbell & Thompson \(2008\)](#)
 - ▶ Inference: block-bootstrap CI for R^2 (blocks of 20 days), Diebold–Mariano (DM) test on squared errors with Newey–West variance ([Diebold & Mariano \(1995\)](#))

The LSTM for Returns

- Input: the last 20 returns, standardised with the mean and standard deviation of the training period
 - ▶ Output: next-day return; loss MSE; Adam; weight decay 10^{-4}
- Trained once on up to 4,000 days before the test period; last 20% of them for early stopping
 - ▶ Five random seeds; the forecast is the average of the five networks
- One hidden layer of 16 units: 1 233 parameters

Training the LSTM: Early Stopping

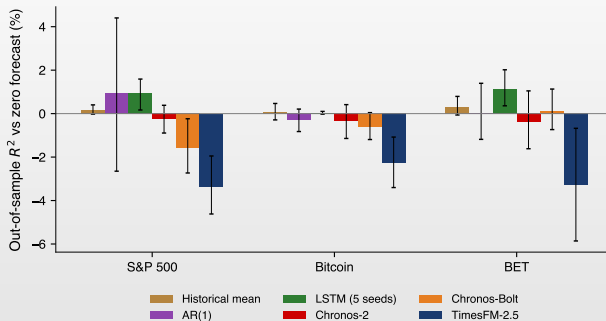


- S&P 500, seed 0: MSE of standardised returns by epoch, on the training and validation parts
- Training stops 26 epochs in; the weights of epoch 1 are kept

Interpretation: Little to Learn

- Both losses barely move; the validation loss is lowest after the first epoch, so early stopping keeps epoch 1
 - ▶ The validation MSE is below the training MSE because the validation years were calmer: the level reflects volatility, not skill
- Early stopping picks a network close to its starting point: forecasts near the mean
 - ▶ Standard deviation of the LSTM forecasts for the S&P 500: 0.047%, against 1.13% for returns
- A healthy sign: a network that forecast large moves would be fitting noise

Out-of-Sample R^2 for Returns



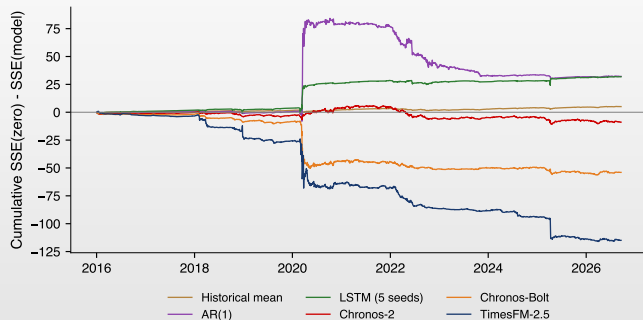
□ R^2 against the zero forecast (%), with 95% block-bootstrap CIs

Interpretation: No Model Beats Zero, Except on the BET

R^2 (%) [DM p]	Historical mean	AR(1)	LSTM (5 seeds)	Chronos-2	Chronos-Bolt	TimesFM-2.5
S&P 500	+0.15 [0.095]	+0.94 [0.639]	+0.93 [0.090]	-0.26 [0.418]	-1.57 [0.078]	-3.35 [0.001]
Bitcoin	+0.07 [0.691]	-0.28 [0.326]	+0.04 [0.243]	-0.35 [0.413]	-0.61 [0.070]	-2.29 [0.001]
BET	+0.30 [0.081]	-0.00 [1.000]	+1.15 [0.014]	-0.39 [0.562]	+0.12 [0.831]	-3.27 [0.070]

- S&P 500 and Bitcoin: no model beats zero significantly; TimesFM-2.5 is significantly **worse** than zero
 - ▶ Foundation models forecast too much movement: the s.d. of TimesFM forecasts on the S&P 500 is 0.141%, three times that of the LSTM (0.047%)
- BET: LSTM R^2 +1.15% (DM $p = 0.014$): a small signal; nesting, multiplicity and stale prices are tested at the end of this section

Cumulative Squared-Error Gains on the S&P 500

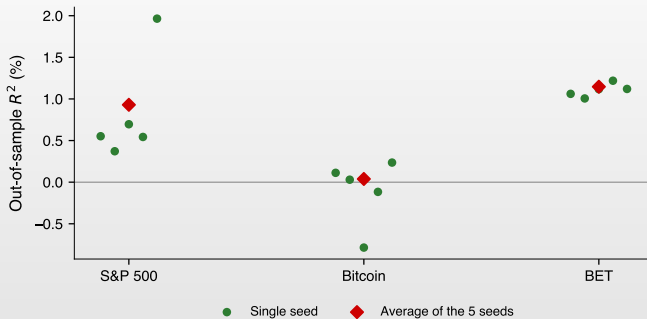


□ $\sum_{s \leq t} [r_s^2 - (r_s - \hat{r}_s)^2]$: rises when the model beats the zero forecast, falls when it loses

Interpretation: Gains Come from a Few Days

- All curves stay near zero in calm years and jump in March 2020
 - ▶ Squared errors are dominated by crisis days: one month decides the ranking
- AR(1) gains in 2020 from the negative autocorrelation of crash days, then gives part back
 - ▶ TimesFM-2.5 loses steadily: its forecasts are too volatile
- Lesson: report the path, not only the final R^2 ; a single episode can create or destroy "predictability"

The Seed Lottery



- Out-of-sample R^2 of five LSTMs that differ only in the random seed, and of their average

Interpretation: Report All Seeds

- ▣ Bitcoin: single seeds range from -0.79% to $+0.24\%$; S&P 500 from $+0.37\%$ to $+1.96\%$
 - ▶ Seed-to-seed variation is as large as the effect itself
- ▣ BET: all 5 seeds positive, between $+1.01\%$ and $+1.22\%$: robust across seeds
 - ▶ Robust across seeds is not robust to multiplicity or to stale prices: next slides
- ▣ Choosing the best seed after looking at test results is backtest overfitting (Chapter 13)

Nested Forecasts: Clark–West, Not Diebold–Mariano

- The zero forecast is nested in the historical mean and in AR(1): set their coefficients to zero
 - ▶ Under H_0 the larger model only adds estimation noise to its squared error: DM rejects too rarely
 - ▶ Clark & West (2007): $\tilde{d}_t = e_{0,t}^2 - \left[e_{1,t}^2 - (\hat{r}_{0,t} - \hat{r}_{1,t})^2 \right]$, one-sided HAC t -test, approximately $N(0, 1)$
- One-sided p , DM versus Clark–West: historical mean 0.048 vs 0.007 (S&P 500), 0.041 vs 0.003 (BET); AR(1) 0.320 vs 0.055 (S&P 500)
 - ▶ BET AR(1), $R^2 = -0.00\%$ and DM $p = 1.000$, is not a degenerate forecast: s.d. 0.074%, average slope 0.046
 - ▶ Squared-error gains of -12.7 in 2020 and +14.7 in 2026 cancel; without these two years $R^2 = -0.11\%$
- Rolling or re-trained models: Giacomini & White (2006) test the forecasting method, estimation window included (Diebold (2015) on the use and abuse of DM)

Many Models, One Test Period: Multiplicity

- 6 models $\times$ 3 markets = 18 tests against zero; the smallest one-sided DM p is the BET LSTM, 0.007
 - ▶ Bonferroni threshold $0.05/18 = 0.0028$; Holm-adjusted p over the 18 tests = 0.130: not significant
- Within the BET family of six models (stationary bootstrap, mean block 20 days)
 - ▶ Romano–Wolf stepwise adjusted p for the LSTM: 0.016 ([Romano & Wolf \(2005\)](#))
 - ▶ SPA test, H_0 : no model beats zero ([Hansen \(2005\)](#)): BET 0.018, S&P 500 0.130, Bitcoin 0.354
 - ▶ SPA refines White's Reality Check ([White \(2000\)](#)): very poor models (TimesFM) no longer dilute the test
- Reading: “some model beats zero on the BET” survives; “one of 18 comparisons is real” does not. Fix the family before looking

Is BET Predictability Real? Non-Synchronous Trading

- Stale prices of thinly traded constituents make index returns positively autocorrelated, even if every price is efficient (Lo & MacKinlay (1990))
 - ▶ First-order autocorrelation since 2016: BET 0.06, S&P 500 -0.14
 - ▶ The LSTM forecast has correlation 0.88 with the previous day's BET return: it mostly learned this autocorrelation
- Tradable check: the BET-tracking ETF Patria-TVBETETF, 2629 common days, same LSTM signal
 - ▶ Same day: R^2 2.88% (BET 1.26%), but the ETF is thinly traded too (autocorrelation 0.07, 4.2% zero-return days)
 - ▶ Executed one day later: R^2 0.57%, Clark–West $p = 0.003$
- Reading: most of the signal is mechanical; a small part survives the execution lag, before transaction costs (Chapter 2)

Section Summary

- ▣ Daily returns of large markets: no deep or foundation model beats the zero forecast
- ▣ Zero-shot point forecasts can be worse than zero: too much movement is predicted
- ▣ BET: the LSTM signal survives within-market tests (Romano–Wolf, SPA) but not the 18-test family, and is mostly stale-price autocorrelation

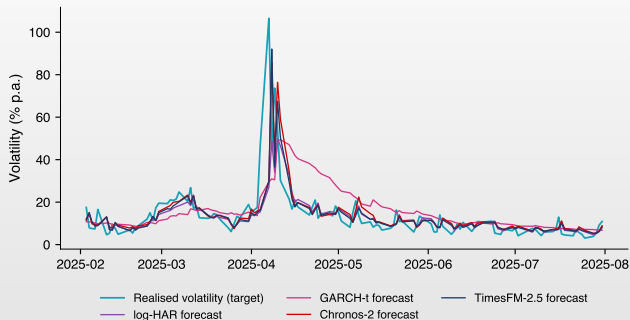
Target, Benchmark and Loss

- ▣ **Realised variance** (RV, Andersen et al. (2003)): sum of squared 5-minute log returns within the trading day
 - ▶ $RV_t = \sum_{j=1}^{78} r_{t,j}^2$ for SPY (regular session, 6.5 hours); average volatility 11.9% p.a.
- ▣ **HAR** (Corsi (2009)): $RV_t = \beta_0 + \beta_d RV_{t-1} + \beta_w \overline{RV}_{t-5:t-1} + \beta_m \overline{RV}_{t-22:t-1} + \varepsilon_t$
 - ▶ log-HAR: the same on log RV, forecast $\exp(\hat{y} + \hat{s}^2/2)$: the mean if the log error is conditionally Normal
- ▣ **QLIKE** loss (Patton (2011)): $L = RV_t/\hat{h}_t - \log(RV_t/\hat{h}_t) - 1$
 - ▶ Minimised by the conditional mean; ranks forecasts as the true variance would when RV is a conditionally unbiased proxy, $E[RV_t | \mathcal{F}_{t-1}] = \sigma_t^2$ (so does MSE)

Design of the Volatility Test

- ▣ SPY realised variance, one-day-ahead forecasts from 11.10.2022 to 18.09.2026 (979 days)
 - ▶ HAR and log-HAR on a rolling 500-day window; random walk: $\hat{h}_t = \text{RV}_{t-1}$
 - ▶ GARCH-t on daily SPY returns, rescaled to RV units by the ratio of average RV to average squared return over 500 days
 - ▶ LSTM on the last 22 values of log RV, retrained every 125 days on 500 days (hidden size 8, three seeds)
- ▣ Foundation models on log RV (context 512): forecast $\hat{h}_t = \int_0^1 \exp(q_u) du$ on the quantile grid
 - ▶ Post-release window: 218 days from 3 November 2025

Forecasts Through the April 2025 Shock

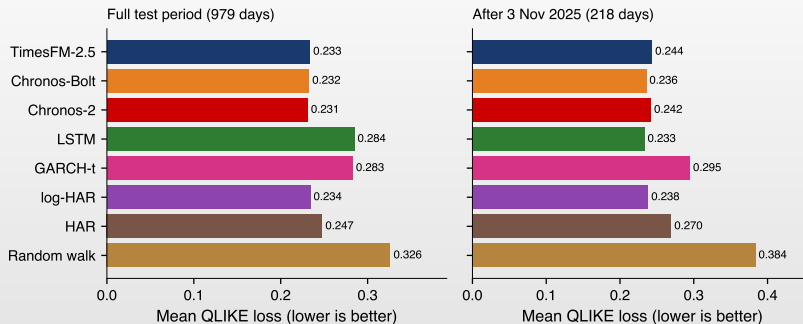


- SPY realised volatility (annualised) and one-day-ahead forecasts, February–July 2025
- Peak: 107% p.a. on 07.04.2025, after the US tariff announcements

Interpretation: Everybody Is One Day Late

- No model anticipates the jump; all of them react the next day
 - ▶ One-day forecasts can only extrapolate the past; shocks come from news
- The foundation models and log-HAR decay back at similar speeds
 - ▶ GARCH-t reacts to the size of daily returns and decays more slowly
- The ranking is decided by the average over about four years: next slide

QLIKE Losses of Eight Volatility Models



□ Left: full test period; right: window after the release of all model weights

 MFM_ch14_realised_vol

Interpretation: No Detectable Difference from log-HAR

Model	QLIKE	DM vs log-HAR	p	MCS p	QLIKE after 3.11.2025
Random walk	0.326	+4.53	<0.001	0.00	0.384
HAR	0.247	+1.30	0.195	0.29	0.270
log-HAR	0.234	—	—	0.95	0.238
GARCH-t	0.283	+3.43	<0.001	0.03	0.295
LSTM	0.284	+1.68	0.094	0.29	0.233
Chronos-2	0.231	-0.62	0.537	1.00	0.242
Chronos-Bolt	0.232	-0.18	0.857	0.95	0.236
TimesFM-2.5	0.233	-0.08	0.938	0.95	0.244

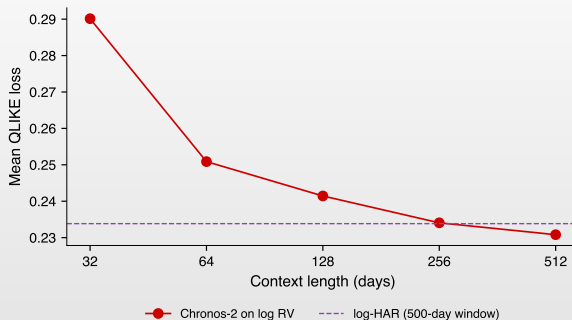
- Chronos-2, Chronos-Bolt, TimesFM-2.5 and log-HAR: DM tests not significant, all in the 90% MCS
- Random walk and GARCH-t are eliminated from the MCS; HAR in levels over-forecasts (mean forecast / mean RV = 1.36)
- After 3 November 2025 the small LSTM has the lowest QLIKE (0.233): over 218 days, rankings are unstable

Median or Mean? A Costly Detail

Chronos-2 and TimesFM-2.5 on log RV	QLIKE	Mean forecast / mean RV
Chronos-2, $\int_0^1 \exp(q_u) du$	0.231	1.02
Chronos-2, $\exp(q_{0.5})$	0.259	0.79
TimesFM-2.5, $\int_0^1 \exp(q_u) du$	0.233	0.94
TimesFM-2.5, $\exp(q_{0.5})$	0.259	0.79

- A theorem, not a curiosity ([Gneiting \(2011\)](#)): QLIKE is consistent for the mean, $\arg \min_h E[L(RV, h) | \mathcal{F}] = E[RV | \mathcal{F}]$
 - ▶ Monotone maps preserve quantiles: $\exp(\text{median of } \log RV)$ is the median of RV, elicited by absolute error, not by QLIKE
 - ▶ Lognormal RV: median/mean = $e^{-s^2/2}$; the median version is significantly worse (DM +3.60 for Chronos-2) and worse than log-HAR (0.234)

How Much Context Does Chronos-2 Need?



□ QLIKE of Chronos-2 on log RV with contexts of 32 to 512 days (same forecast days)

 MFM_ch14_realised_vol

Interpretation: Long Memory Needs Long Context

- ▣ QLIKE falls from 0.290 with 32 days to 0.231 with 512 days
 - ▶ With less than about 256 days, Chronos-2 is worse than log-HAR (0.234)
- ▣ Volatility has long memory: months of history tell the model the slow-moving level
 - ▶ HAR encodes the same idea explicitly, through its 22-day average
- ▣ Context length is a modelling choice to report and to test, like a window length

Section Summary

- ▣ Zero-shot foundation models reach QLIKE losses close to a well-specified log-HAR; the DM tests detect no difference (not proof of equality)
- ▣ They are not significantly better, before or after the release of their weights
- ▣ No conditional difference detected with the chosen instruments: Giacomini–White test $(1, d_{t-1})$ against log-HAR, $p = 0.760$ (Chronos-2), 0.935 (TimesFM-2.5)
- ▣ Use the mean, not the median, and a long context; always compare with HAR

Design of the Tail-Risk Test

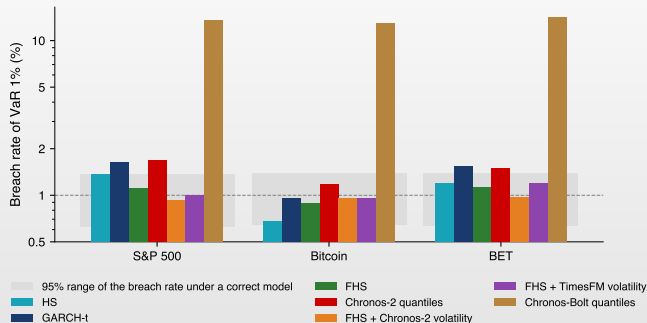
- $\text{VaR}_\alpha = -q_\alpha(r_{t+1} \mid \mathcal{F}_t)$, the loss exceeded with probability α ; $\text{ES}_\alpha = -\frac{1}{\alpha} \int_0^\alpha q_u du$
 - ▶ Levels: VaR 1% (coverage tests) and the pair (VaR 2.5%, ES 2.5%) for the FZ0 loss (Fissler & Ziegel (2016), Patton, Ziegel & Chen (2019))
- Classical models on a 1,000-day window: HS, GARCH(1,1)-t, FHS (McNeil & Frey (2000)), re-estimated every 20 days
- Foundation models, two ways
 - ▶ **Raw**: Chronos-2 quantiles of tomorrow's return (context: 512 returns)
 - ▶ **Hybrid**: scale $\hat{\sigma}_t = \text{forecast of } |r_t|$ (Chronos-2: grid mean; TimesFM: median); FHS on $r_t/\hat{\sigma}_t$, unaffected by a constant rescaling of $\hat{\sigma}_t$

Scoring (VaR, ES) Jointly: the FZ0 Loss

- ES alone is not elicitable, the pair $(\text{VaR}_\alpha, \text{ES}_\alpha)$ is (Fissler & Ziegel (2016)): it has consistent joint scoring functions
 - ▶ FZ0 (Patton, Ziegel & Chen (2019)), loss $L_t = -r_t$, $v = \text{VaR}_\alpha$, $e = \text{ES}_\alpha$:

$$\text{FZ0} = \frac{1\{L_t > v\}(L_t - v)}{\alpha e} + \frac{v}{e} + \log e - 1$$
 - ▶ Domain: loss-convention ES $e > 0$ (the logarithm and the ratios need it); here $\alpha = 2.5\%$
- Lower is better; the expected FZ0 is minimised by the true (VaR, ES) pair
 - ▶ Penalises breaches (first term) and wide forecasts (the $\log e$ term): calibration and sharpness in one number
 - ▶ Models are compared by DM tests on FZ0 differences and by the MCS (Hansen, Lunde & Nason (2011))

VaR 1% Breach Rates



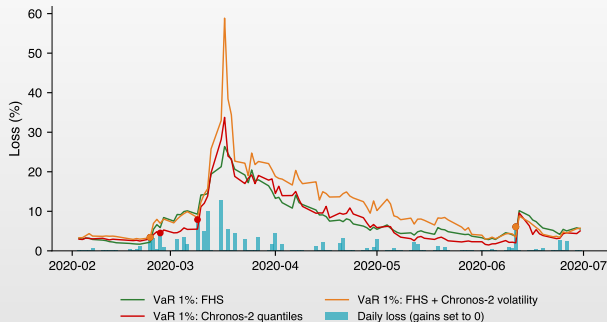
- Share of days with a loss above VaR 1% (log scale); grey band: 95% range of the breach rate if the model were correct
- Chronos-Bolt: its lowest quantile (10%) used as "VaR 1%"

Interpretation: Raw Quantiles Are Too Narrow

Breaches [Kupiec p]	HS	GARCH-t	FHS	Chronos-2 quantiles	FHS + Chronos-2 vol.	FHS + TimesFM vol.
S&P 500 (26.9)	37 [0.065]	44 [0.002]	30 [0.559]	45 [0.001]	25 [0.705]	27 [0.989]
Bitcoin (28.2)	19 [0.065]	27 [0.822]	25 [0.539]	33 [0.374]	27 [0.822]	27 [0.822]
BET (26.8)	32 [0.322]	41 [0.010]	30 [0.536]	40 [0.016]	26 [0.884]	32 [0.322]

- In brackets after the market: the expected number of breaches, 1% of the days
- Raw Chronos-2 quantiles: too many breaches on the S&P 500 (1.67%); Chronos-Bolt "VaR 1%": 13.5%
 - ▶ Hybrid FHS with foundation-model volatility: breach rates close to 1% in all three markets (average coverage only; clustering is tested below)

The COVID-19 Crash: VaR 1% of Three Models



□ S&P 500, February–June 2020 (104 days): daily losses and VaR 1% forecasts; dots: breaches

Interpretation: Speed of Reaction

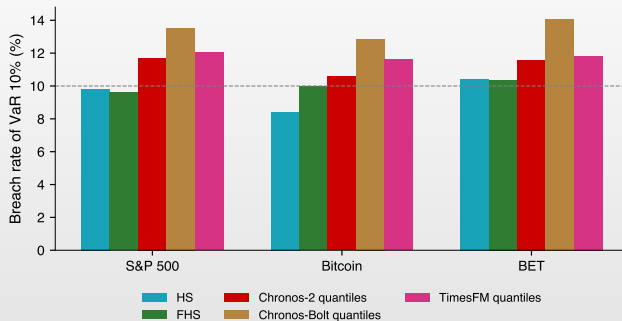
- Breaches in the 104 days: FHS 2, raw Chronos-2 4, FHS with Chronos-2 volatility 2, HS 12
 - ▶ All models are hit in the first days of the crash: none anticipates it
- Chronos-2 reads the context in the first days, so its quantiles widen fast
 - ▶ Chronos-2 standardises the context by its mean and standard deviation, then applies $\sinh^{-1}$ (Ansari et al. (2025)); on 16.03.2020 the crash raised the 512-day standard deviation by 9.7% (mean absolute scale of Chronos: 3.2%, Seminar A4)
 - ▶ So the widening mixes the scale change and the reading of the crash days; only an ablation could separate them
- The hybrid overshoots: VaR 1% of 58.9% on 17.03.2020, against 26.4% for FHS and a largest loss of 12.8%
 - ▶ Not a unit error: after the crash day Chronos-2 forecast a mean $|r|$ of 12.5% for the next day, then FHS multiplies it by the 1% quantile of $r/\hat{\sigma}$ (4.7)
 - ▶ The filter extrapolates a one-day spike as a new level: VaR 4.6 times the largest loss, a high capital cost that the FZ0 loss penalises

Backtesting Beyond Kupiec: Clustering and the DQ Test

p : indep. / DQ	HS	GARCH-t	FHS	Chronos-2 raw	FHS + Chronos-2	FHS + TimesFM
S&P 500	0.001 / <0.001	0.205 / 0.001	0.045 / 0.021	0.223 / <0.001	0.021 / 0.001	0.029 / <0.001
Bitcoin	0.611 / 0.760	0.261 / 0.463	0.220 / 0.092	0.404 / 0.034	0.026 / 0.004	0.261 / 0.182
BET	0.399 / <0.001	0.258 / 0.082	0.409 / 0.650	0.270 / 0.026	0.475 / <0.001	0.379 / <0.001

- Independence ([Christoffersen \(1998\)](#)): first-order Markov chain of breaches, $LR \sim \chi_1^2$; conditional coverage = Kupiec + independence, χ_2^2
- Dynamic quantile (DQ, [Engle & Manganelli \(2004\)](#)): regress $I_t - \alpha$ on a constant, four lagged hits and VaR_t ; Wald statistic $\sim \chi_6^2$
- S&P 500: the hybrids have the right breach rate but fail independence and DQ; every model fails DQ at 5%
 - BET: FHS passes both tests, the hybrids fail DQ; right on average is not right conditionally

Where Deciles Suffice: VaR 10%

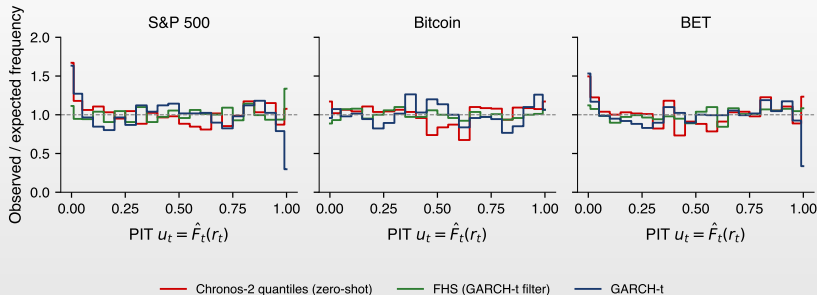


- Breach rate of the 10% quantile (target 10%) for HS, FHS and the three foundation models

Interpretation: Even the 10% Quantile Is Too High

- ▣ S&P 500: HS 9.8%, FHS 9.6%; Chronos-2 11.7%, Chronos-Bolt 13.5%, TimesFM-2.5 12.0%
 - ▶ Bitcoin: Chronos-2 10.6%, Chronos-Bolt 12.8%, TimesFM-2.5 11.6%; BET: 11.6%, 14.1%, 11.8%
- ▣ All three foundation models breach their 10% quantile too often: predictive distributions too narrow for returns
 - ▶ A plausible reason: pretraining corpora are dominated by smoother series (energy, traffic, web) with lighter tails than returns
- ▣ Used as volatility filters, the same models inherit the tails from the data, through FHS

Calibration of the Whole Distribution: the PIT



- Probability integral transform (PIT) $u_t = \hat{F}_t(r_t)$: i.i.d. uniform on $(0, 1)$ if the predictive distribution is correct (Diebold, Gunther & Tay (1998))
- Observed over expected frequency in the 22 intervals of the Chronos-2 grid (1%, 5%, 10%, ..., 99%); 1 = perfect calibration

Interpretation: Berkowitz Test and Quantile Score

Market	Berkowitz p : Chronos-2	FHS	Below $q_{1\%}$ (C2)	QS: C2 / FHS	DM [p]
S&P 500	0.019	0.599	1.67%	0.510 / 0.506	+3.06 [0.002]
Bitcoin	0.595	0.306	1.17%	1.526 / 1.529	-0.90 [0.369]
BET	0.045	0.436	1.50%	0.465 / 0.460	+3.48 [<0.001]

- Berkowitz (2001): LR test of $\mu = 0, \sigma = 1$ for $z_t = \Phi^{-1}(u_t)$, censored outside [1%, 99%] where the grid is silent
 - Chronos-2 rejected on the S&P 500 ($\hat{\sigma} = 1.03$) and the BET; FHS never rejected
- Quantile score (QS) = $\frac{2}{K} \sum_k (1\{r_t < q_{u_k}\} - u_k)(q_{u_k} - r_t)$: an equally weighted grid analogue of the continuous ranked probability score (CRPS, Gneiting & Raftery (2007))
 - Chronos-2 worse than FHS in the centre and in the tails on the S&P 500 (DM +2.51, +3.65) and the BET; tie on Bitcoin

Comparing (VaR 2.5%, ES 2.5%) Forecasts: the MCS

HS	0.140	0.343	0.015
GARCH-t	0.187	0.597	0.341
FHS	1.000	0.597	1.000
Chronos-2 quantiles	0.140	0.641	0.239
FHS + Chronos-2 volatility	0.140	1.000	0.341
FHS + TimesFM volatility	0.183	0.704	0.202
	S&P 500	Bitcoin	BET

□ p -values of the 90% model confidence set ([Hansen, Lunde & Nason \(2011\)](#)) for the FZ0 loss; green: kept, red: eliminated

Interpretation: FHS Remains the Reference

FZ0 [MCS p]	HS	GARCH-t	FHS	Chronos-2 quantiles	FHS + Chronos-2 vol.	FHS + TimesFM vol.
S&P 500	1.309 [0.14]	1.026 [0.19]	0.999 [1.00]	1.147 [0.14]	1.084 [0.14]	1.059 [0.18]
Bitcoin	2.237 [0.34]	2.192 [0.60]	2.192 [0.60]	2.181 [0.64]	2.151 [1.00]	2.160 [0.70]
BET	1.230 [0.01]	1.001 [0.34]	0.963 [1.00]	1.032 [0.24]	1.007 [0.34]	1.027 [0.20]

- S&P 500: FHS has the lowest FZ0 loss; hybrids have the right breach rate but clustered breaches and a larger VaR (3.20% vs 2.78% on average)
- Bitcoin: the hybrid FHS with Chronos-2 volatility has the lowest FZ0 loss, but the differences are not significant
- The MCS eliminates no model on the S&P 500 and Bitcoin, only HS on the BET; pairwise DM tests are sharper: FHS beats the Chronos-2 hybrid on the S&P 500 (DM +2.65, $p = 0.008$) and on the BET (DM +2.36, $p = 0.018$)

After the Release of the Weights

Breaches after 3.11.2025	HS	GARCH-t	FHS	Chronos-2 quantiles	FHS + Chronos-2 vol.	FHS + TimesFM vol.
S&P 500 (220 days)	0	2	2	2	2	2
Bitcoin (320 days)	3	3	4	4	3	4
BET (217 days)	4	4	4	5	1	2

- Expected number of breaches: 2.2 (S&P 500), 3.2 (Bitcoin), 2.2 (BET)
 - ▶ No pretraining contamination is possible here, but with two or three expected breaches the tests have little power
 - ▶ Low power is not zero power: HS has 0 breaches on the S&P 500, and $x = 0$ is in the Kupiec rejection region at $T = 220$ (Seminar A8)
- For tail risk, the post-release window has low power against modest coverage errors (next slide) and a size far from 5%: a non-rejection does not establish correct calibration

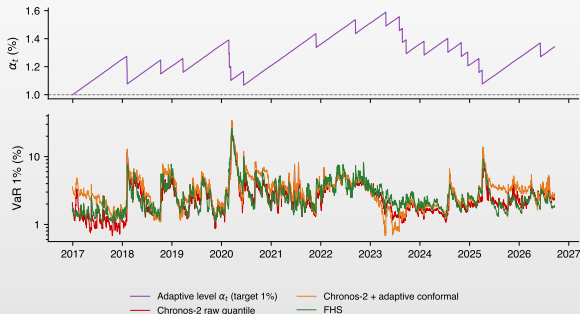
Power of the Post-Release Window

- Two-sided Kupiec LR test at 5%, VaR 1%; alternative: a true breach rate of 1.67% (raw Chronos-2 on the S&P 500); exact binomial power
 - ▶ $T = 220$ days: 2.2 expected breaches, power 0.19; actual size 0.13 instead of 0.05, because of discreteness
 - ▶ $T = 1,000$: power 0.50; $T = 2,693$ (our full sample): 0.87
 - ▶ 80% power first reached at about 2 130 days (8.4 years), and for good from 2 320 days
- Even against a true rate of 2%: power 0.29 at $T = 220$; 80% needs about 1 020 days
- ES comparisons are harder still: finite-sample precision limits for ES forecast comparisons ([Pele & Mazurencu-Marinescu-Pele \(2026\)](#)); ES is backtestable jointly with VaR ([Nolde & Ziegel \(2017\)](#))

Recalibrating Quantiles: Adaptive Conformal Inference

- Score $E_t = \hat{q}_{\alpha,t} - r_t$; corrected quantile $\hat{q}_{\alpha,t} - \hat{Q}_{1-\alpha_t}$, $\hat{Q}$: the $\lceil (1 - \alpha_t)(250 + 1) \rceil$ -th smallest of $E_{t-250}, \dots, E_{t-1}$, $+\infty$ if that rank exceeds 250
 - ▶ Adaptive level ([Gibbs & Candès \(2021\)](#)): $\alpha_{t+1} = \alpha_t + \gamma(\alpha - \text{err}_t)$, $\gamma = 0.001$ ($\gamma/\alpha = 0.1$); $\gamma = 0$: plain split conformal
 - ▶ With $\gamma = 0.005$ the level would reach the $+\infty$ region: infinite VaR on 24 (S&P 500), 100 (Bitcoin), 270 (BET) days
- Guarantee for any sequence, no model needed: $\left| \frac{1}{T} \sum_t \text{err}_t - \alpha \right| \leq \frac{\max(\alpha_1, 1 - \alpha_1) + \gamma}{\gamma T}$
 - ▶ A long-run statement: with $T = 2443$ and $\gamma = 0.001$ the bound is 0.41, uninformative for a 1% rate
 - ▶ Without exchangeability, weighted conformal bounds the coverage gap by the distribution drift ([Barber et al. \(2023\)](#))
- ES 2.5%: Chronos-2 ES shifted by the VaR 2.5% correction; below 1% a GPD tail beyond $q_{0.01}$ ([McNeil & Frey \(2000\)](#)) is the consistent alternative (Seminar A6)

Adaptive Conformal VaR 1% on the S&P 500



- Top: adaptive level α_t (target 1%); bottom: VaR 1% of raw Chronos-2, Chronos-2 with adaptive conformal correction, FHS (log scale)

Interpretation: Coverage Restored, Sharpness Not

After 250 calibration days:				
breaches [cc p / DQ p]	FHS	Chronos-2 raw	FHS + C2 vol.	Chronos-2 + ACI
S&P 500 (24.4)	28 [0.097 / 0.014]	42 [0.002 / <0.001]	23 [0.058 / <0.001]	21 [0.307 / 0.049]
Bitcoin (25.7)	24 [0.449 / 0.108]	31 [0.410 / 0.026]	26 [0.087 / 0.004]	24 [0.449 / 0.191]
BET (24.2)	26 [0.709 / 0.727]	36 [0.048 / 0.184]	23 [0.776 / 0.011]	25 [0.762 / 0.032]
FZ0, S&P 500 [DM p]	1.013	1.142 [0.009]	1.087 [0.022]	1.205 [0.001]

- Adaptive conformal inference (ACI) brings the breach rate near 1% and conditional coverage is not rejected in any market; DQ still rejects on the S&P 500 and the BET, passes on Bitcoin
- But its VaR is wider (S&P 500: 3.36% vs 2.85% for FHS) and its FZ0 loss significantly higher on the S&P 500 and the BET
- Proper scoring rules reward sharpness subject to calibration ([Gneiting & Raftery \(2007\)](#)): calibration is necessary, not sufficient

Section Summary

- ▣ Raw foundation-model quantiles are too narrow for returns, even at 10%; Chronos-Bolt and TimesFM cannot reach 1% at all
- ▣ As volatility filters inside FHS they give near-target breach rates but clustered breaches on the S&P 500; the Chronos-2 hybrid has a significantly higher FZ0 loss than GARCH-based FHS on both the S&P 500 and BET (the TimesFM hybrid only on the BET)
- ▣ Conformal recalibration brings coverage near target without a GARCH filter, but not dynamic calibration (DQ) or sharpness; the post-release window has little power
- ▣ None of them replaces GARCH-based FHS on the evidence so far

Case Study: Is Complexity a Virtue in Return Prediction?

- **Question** (Kelly, Malamud & Zhou (2024)): must a return-prediction model have far fewer parameters P than observations T , or can models with complexity $c = P/T > 1$ do better out of sample?
 - ▶ Setting (eqs. 1–2): $R_{t+1} = f(G_t) + \varepsilon_{t+1}$, f unknown, approximated by a wide one-hidden-layer network $\sum_{i=1}^P S_{i,t} \beta_i$ (universal approximation, Hornik, Stinchcombe & White (1989))
- **Link to Test 1**: our networks do not beat the zero forecast in R^2 ; the paper asks whether a negative out-of-sample R^2 means no economic value
 - ▶ Economic value: Sharpe ratio of the market-timing return $\pi_t R_{t+1}$, with position $\pi_t = \hat{\beta}' S_t$ (eq. 6)
- **Theory** (Sections III–IV): for a misspecified model with ridge shrinkage, the expected out-of-sample R^2 and timing Sharpe ratio rise with c
 - ▶ Near $c = 1$ (the interpolation boundary) least squares fits the training data exactly and its variance explodes; beyond it, performance recovers
- **Why it matters**: a theoretical case for the large models of this chapter, against the parsimony rule of classical econometrics

Case Study: Data and Method

- **Data** (Section V.A): monthly excess return of the CRSP value-weighted index; the 15 predictors of [Welch & Goyal \(2008\)](#) (14 variables and the lagged market return), 1926–2020
 - ▶ Returns scaled by their trailing 12-month volatility, predictors by an expanding-window standard deviation; 36 months of warm-up, so the sample starts in 1930
- **Random Fourier features** (eq. 20, [Rahimi & Recht \(2007\)](#)): $S_{i,t} = (\sin(\gamma \omega_i' G_t), \cos(\gamma \omega_i' G_t))'$, $\omega_i \sim N(0, I)$, $\gamma = 2$
 - ▶ A wide two-layer network whose first-layer weights ω_i are drawn at random and fixed; only the output weights β are estimated
- **Ridge** (Section II.A): $\hat{\beta}(z) = (zI + T^{-1} \sum_t S_t S_t')^{-1} T^{-1} \sum_t S_t R_{t+1}$, no intercept; $z \rightarrow 0^+$: the ridgeless (pseudo-inverse) solution
- **Design** (Section V.C): rolling windows of $T = 12, 60, 120$ months; P from 2 to 12 000; $\log_{10} z = -3, \dots, 3$; results averaged over 1 000 independent draws of the features
- **Evaluation**: out-of-sample $R^2 = 1 - \text{Var}(\text{error})/\text{Var}(R)$; annualised Sharpe ratio; alpha and information ratio (alpha over residual volatility) from a regression of the timing return on the market
 - ▶ Benchmark: the linear "kitchen sink" regression on all 15 predictors (Section V.D)

Case Study: Key Results

Model ($T = 12$)	R^2 (%)	Sharpe [t]	Inf. ratio [t]	Max loss	Skewness
Linear, $z = 0^+$	< -100	-0.11 [-1.0]	-0.16 [-1.6]	98.5	-0.9
Linear, $z = 10^3$	-3.8	0.46 [4.4]	0.33 [3.1]	2.4	-0.1
RFF, $c = 1\,000$, $z = 10^3$	0.6	0.47 [4.5]	0.31 [2.9]	1.2	2.5

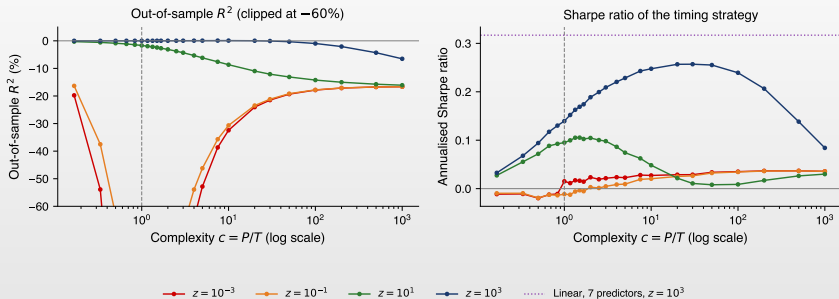
Source: Table I, Panel A, of the paper; information ratio against the market; maximum monthly loss in standard deviations

- ▣ Ridgeless kitchen sink ($P = 15 > T = 12$): R^2 below -100% , Sharpe ratio -0.11
 - ▶ Reading: at the interpolation boundary forecasts explode, reproducing the negative verdict of [Welch & Goyal \(2008\)](#)
- ▣ Same predictors with $z = 10^3$: $R^2 = -3.8\%$, yet Sharpe ratio 0.46 ($t = 4.4$)
 - ▶ Reading: R^2 penalises the scale of forecasts; timing profits depend on their correlation with returns (footnote 5)
- ▣ RFF, $c = 1\,000$: information ratio 0.26 ($t = 2.5$) against the best linear model; with $T = 60$ and 120 , 0.25 and 0.24 against the market ($t = 2.3$ and 2.2 ; Panels B–C)
 - ▶ Reading: nonlinearity extracts more from the same predictors, with a smaller maximum loss and positive skewness
- ▣ Positions (Figure 10): almost always long; cut before 14 of 15 NBER recessions, out of sample
 - ▶ Reading: the model learns the sign restriction of [Campbell & Thompson \(2008\)](#) without being given it

Case Study: Replicate, Limits, Extend

- **Replicate** with the course data (Seminar 14, C3): monthly SPY total returns from 1993 (data/market) in excess of the 3-month Treasury bill rate (FRED TB3MS)
 - ▶ Predictors of [Welch & Goyal \(2008\)](#) available: `tbl`, `lty`, `tms`, `dfy`, `inf1` (FRED), `svar` from daily S&P 500 returns, the lagged excess return: 7 of the 15
 - ▶ Data limitation: the valuation ratios (`dp`, `dy`, `ep`, `de`, `b/m`), `ntis` and the bond returns `ltr`, `dfr` are not in the course data; about 30 years instead of 90
 - ▶ Method unchanged: $\gamma = 2$, $T = 12$, $P \leq 12\,000$, $\log_{10} z = -3, \dots, 3$, 1 000 draws
- **Limits** stated by the authors
 - ▶ Theory for linear ridge models on fixed random features and a single asset; a panel with cross-asset covariances is left open (Introduction, footnote 2)
 - ▶ Predictability spread evenly across features (Assumption 4); sparse LASSO-type models left to future research (footnote 22)
 - ▶ No optimal degree of nonlinearity γ ; average timing returns about half as large in 1975–2020 as in 1930–1974 (Section V.F)
- **Open extension**
 - ▶ Learned instead of random features: the internal representations (embeddings) of a pretrained foundation model such as Chronos-2 as S_t ; is complexity still a virtue?
 - ▶ A panel of course markets (S&P 500, BET, Bitcoin), and the power of a 30-year test for an information ratio of 0.3

Case Study: Complexity on Course Data

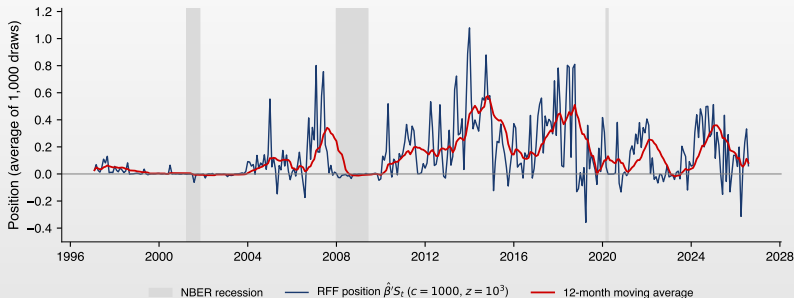


- Figures 7–8 of Kelly, Malamud & Zhou (2024) on course data: SPY excess returns, 7 predictors, 354 test months (February 1997–August 2026); $T = 12$, $\gamma = 2$, averages over 1000 draws
- Dashed line: interpolation boundary $c = 1$; dotted line: linear kitchen sink with $z = 10^3$; left panel cut at -60%

Interpretation: a Virtue up to a Point

- Nearly ridgeless ($z = 10^{-3}$): $R^2 = -1.842\%$ at $c = 1$, recovering to -16.6% at $c = 1\,000$: the double descent of Figure 7
 - ▶ Linear kitchen sink with $z = 0^+$ ($c = 7/12$): $R^2 = -1.540\%$, Sharpe ratio -0.06 , as in Table I
- $z = 10^3$: Sharpe ratio 0.14 at $c = 1$, 0.26 at $c = 30$, then 0.08 at $c = 1\,000$ ($t = 0.5$; $R^2 = -6.5\%$)
 - ▶ Reading: complexity helps up to $c \approx 30$ but not up to $c = 1\,000$ with 7 of the 15 predictors and 29.5 years
- Linear, $z = 10^3$: $R^2 = 0.26\%$, Sharpe ratio 0.32 ($t = 1.7$), information ratio 0.25 ($t = 1.3$) against SPY
 - ▶ RFF with $c = 1\,000$: information ratio -0.17 ($t = -0.9$); in 1997–2026 the shrunk linear model is not beaten, unlike 1930–2020 in the paper
- Power: with 29.5 years, $t \approx 0.31\sqrt{29.5} \approx 1.7$; even the effect in the paper would not be significant at 5% here

Case Study: Timing Positions and Recessions



- Figure 10 of [Kelly, Malamud & Zhou \(2024\)](#) on course data: position $\hat{\beta}' S_t$ of RFF with $c = 1000$, $z = 10^3$, averaged over 1 000 draws, February 1997–August 2026
- Shaded: NBER recessions (FRED USREC); red line: 12-month moving average

Interpretation: Long Most of the Time, Flat in Crises

- Long in 72% of months, average position 0.13
 - ▶ Reading: as in Figure 10, the model learns a positive equity premium without a sign restriction
- Average position 0.07 in the 6 months before the 3 recessions (2001, 2008, 2020) and -0.01 during them, against 0.15 in other months
 - ▶ Reading: exposure near zero in 2000–2003 and from early 2008 to 2009, the recession pattern of the paper
 - ▶ The 2020 recession, a pandemic shock, is not anticipated by these predictors
- Three recessions are a description, not a test: the paper has 15 recessions and 90 years

An Open Question: Skill or Memory?

- Test 2: zero-shot Chronos-2 is not distinguishable from log-HAR on SPY realised variance (QLIKE 0.231 vs 0.234, DM $p = 0.537$)
- Rahimikia, Ni & Wang (2025): on daily excess returns across global markets, off-the-shelf foundation models do poorly zero-shot and after fine-tuning; models pretrained from scratch on financial data gain
- Lopez-Lira, Tang & Zhu (2025): once a model has seen the realised values, its counterfactual forecasting skill is not identified; detected memorisation is only a lower bound (shown for LLMs; the logic applies to any pretrained forecaster)
- Open: is the parity learned structure (long memory, mean reversion) or recall of the test period, and can a short post-release window tell?
 - ▶ $d_t = L(\hat{h}_t^{FM}, RV_t) - L(\hat{h}_t^{\log HAR}, RV_t)$; memorisation implies $E[d_t | \text{post}] > E[d_t | \text{pre}]$; H_0 : equality

The Discovery Loop, Step by Step (1/2)

- **Literature:** "Find evaluations of time-series foundation models on financial returns or volatility since 2024: models, data, loss, benchmark, and whether the test period overlaps the pretraining corpus."
 - ▶ Check every arXiv identifier and read the corpus documentation yourself; Semantic Scholar, Google Scholar, Elicit help with the search
- **Hypotheses:** "List predictions that separate memorisation from skill for a zero-shot volatility forecaster: pre vs post release, never-public series, simulated series with the same dynamics."
 - ▶ Skill predicts equal gains on a simulated log-HAR path; memory predicts gains only on public series before the cutoff
- **Data and code:** "From `ch14_rv.csv`, compute the QLIKE differential Chronos-2 minus log-HAR, test pre vs post 3 November 2025 with HAC errors, and report the minimum detectable effect at 80% power."
 - ▶ Any assistant (Claude, ChatGPT, Gemini, GitHub Copilot); first reproduce the chapter's QLIKE of log-HAR, 0.234

The Discovery Loop, Step by Step (2/2)

- **Robustness**, fixed in advance: "Design a pre-registered test with more power: which cross-section of series, which variance proxy, which panel DM test, and how many post-release days."
 - ▶ Placebo: a fake release date one year earlier must show no break; simulated series cannot be memorised
- **Critique**: "Attack this claim: no memorisation, because the post-release DM test is insignificant. What is wrong with the inference?"
 - ▶ Expect: low power, absence of evidence, and different regimes: the April 2025 tariff shock lies in the pre-release sample, the post-release window is calmer
- **Report**: model versions and weight release dates, context length, mean or median, every seed and window tried

What the Researcher Must Check

- **Pretraining overlap:** a public series may sit in the corpus
 - ▶ Test: post-release window, simulated or private series; record the release date of the weights
- **Low power sold as a null result:** "not significant" over a few hundred days
 - ▶ Test: report the minimum detectable effect (MDE) and the CI of $E[d_t]$, not only p
- **Wrong functional:** the median of log RV used as a mean forecast
 - ▶ Test: mean forecast / mean RV should be close to 1 (Chronos-2 median: 0.79)
- **Leakage in your own code:** HAR averages or scalings that include day t (Seminar B8)
 - ▶ Test: recompute the forecast for t from data truncated at $t - 1$; it must not change
- **Selection:** contexts, seeds and models chosen after seeing the test
 - ▶ Test: log every trial; compare the whole set with the MCS ([Hansen, Lunde & Nason \(2011\)](#))

Mini-Case: a Test Without Power

- A typical assistant proposal: "the parity of Chronos-2 comes from memorised data, so its edge must vanish after release". We test it on the saved forecasts
- Mean QLIKE differential d_t : -0.0050 before 3 November 2025 (761 days) vs +0.0038 after (218 days)
- Change +0.0088, HAC $t = 1.00$ ($p = 0.32$): the direction memorisation predicts, but H_0 is not rejected
- Power of the test of β_1 (5% two-sided, 80% power): $MDE = (z_{0.975} + z_{0.8}) se_{HAC}(\hat{\beta}_1) = 2.80 \times 0.0088 = 0.0247$
 - ▶ 10.6% of the log-HAR QLIKE, 4.9 times the pre-release gap
 - ▶ Even an infinitely long post-release window leaves an MDE of 0.0169 (3.4 times the gap): the pre-release mean is itself estimated from 761 days
- Reading: the window cannot separate memory from skill; "no memorisation" is not a finding. Power must come from the cross-section, not from waiting

 ai_discovery_case.py

Your Turn: a Project Seed

- Question: does zero-shot skill for daily variance survive where memorisation is impossible: after release, on simulated series, on less-covered markets?
- Replicate first: the chapter's QLIKE of log-HAR (0.234) and Chronos-2 (0.231), DM $p = 0.537$, and the 218-day post-release window
- Deliverables
 - ▶ A pre-registered power calculation; a panel of series from the course data with a range-based variance proxy and a proxy-robust loss ([Patton \(2011\)](#))
 - ▶ Pre/post tests with placebo release dates; the MCS across models; a referee-style critique and your reply
- Log each prompt that shaped a decision in AI_USE.md; each invented reference, leak or wrong number caught in AI_ERRORS.md, with the test that caught it ([templates](#))

Pitfalls Checklist

- ▣ **Leakage:** pretraining overlap, scaling with full-sample statistics, shuffled validation windows
 - ▶ Seminar B8: HAR averages that include the forecast day cut QLIKE by a quarter, a gain no honest model achieves
- ▣ **Weak benchmarks:** compare with zero (returns), log-HAR (volatility), FHS (tails), not with a random walk
- ▣ **Selection:** seeds, context lengths, architectures chosen after seeing the test period
- ▣ **Wrong functional:** medians used as means; deciles used as 1% quantiles
- ▣ **Wrong inference:** DM on nested models, no multiplicity control, coverage without independence, "not rejected" without power

When to Use What

Target	Reference model	Where deep learning can help
Daily index returns	Zero forecast	Less efficient markets (BET); many assets, many features
Realised volatility	log-HAR	Zero-shot forecasts for thousands of series without estimation
VaR 1%, ES 2.5%	FHS (GARCH filter)	Volatility filter in FHS or conformal recalibration

- Foundation models are cheap to try: no estimation, one line of code, a CPU is enough
- Their value is breadth (many series, no tuning), not accuracy on a single well-studied series

Further Reading

- Gu, Kelly & Xiu (2020): machine learning for the cross-section of expected returns
- Kelly, Malamud & Zhou (2024): why very large models can predict returns better
- Ansari et al. (2024): Chronos, time series read as a language
- Crypto applications: liquidity forecasting Agakishiev, Härdle & Pele (2025), early warning for “zombie crypto-assets” efBSW, statistical classification of crypto-assets efPele
- Textbooks: Franke, Härdle & Hafner (2019) and Härdle, Chen & Overbeck (2017) for the statistics of financial markets behind the benchmarks

Key Takeaways

- ▣ LSTMs solve the vanishing gradient through the forget gate; Transformers avoid it through attention
- ▣ Foundation models forecast zero-shot from a context window and return quantiles
- ▣ Returns: nothing beats zero on large markets; volatility: foundation models not distinguishable from log-HAR; tails: use them only as volatility filters
- ▣ Evaluate every new model against a strong benchmark and after the release of its weights, with inference that matches the design: Clark–West for nested models, SPA or Romano–Wolf for many models, PIT, quantile scores and DQ for distributions

References (1/7)

- ▣ Agakishiev, I., Härdle, W.K., Pele, D.T. (2025). [Cryptocurrency liquidity forecasting](#). In *Springer Handbooks of Computational Statistics*, 41–60. Springer.
- ▣ Aksu, T., Woo, G., Liu, J., Liu, X., Liu, C., Savarese, S., Xiong, C., Sahoo, D. (2024). [GIFT-Eval: a benchmark for general time series forecasting model evaluation](#). arXiv:2410.10393.
- ▣ Andersen, T.G., Bollerslev, T., Diebold, F.X., Labys, P. (2003). [Modeling and forecasting realized volatility](#). *Econometrica*, 71(2), 579–625.
- ▣ Ansari, A.F., Shchur, O., Küken, J., Auer, A., et al. (2025). [Chronos-2: from univariate to universal forecasting](#). arXiv:2510.15821.
- ▣ Ansari, A.F., Stella, L., Turkmen, C., Zhang, X., et al. (2024). [Chronos: learning the language of time series](#). *Transactions on Machine Learning Research*; arXiv:2403.07815.
- ▣ Barber, R.F., Candès, E.J., Ramdas, A., Tibshirani, R.J. (2023). [Conformal prediction beyond exchangeability](#). *Annals of Statistics*, 51(2), 816–845.
- ▣ Będowska-Sójka, B., Wójcik, P., Pele, D.T. (2026). [Early warning systems for cryptocurrency markets: predicting 'zombie' assets using machine learning](#). *North American Journal of Economics and Finance*, 81, 102543.
- ▣ Bengio, Y., Simard, P., Frasconi, P. (1994). [Learning long-term dependencies with gradient descent is difficult](#). *IEEE Transactions on Neural Networks*, 5(2), 157–166.
- ▣ Berkowitz, J. (2001). [Testing density forecasts, with applications to risk management](#). *Journal of Business & Economic Statistics*, 19(4), 465–474.
- ▣ Brini, A. (2026). [Forecasting realized volatility with time series foundation models: a comparison with econometric benchmarks](#). arXiv:2607.05291.

References (2/7)

- ▣ Campbell, J.Y., Thompson, S.B. (2008). Predicting excess stock returns out of sample: can anything beat the historical average? *Review of Financial Studies*, 21(4), 1509–1531.
- ▣ Christoffersen, P.F. (1998). Evaluating interval forecasts. *International Economic Review*, 39(4), 841–862.
- ▣ Clark, T.E., West, K.D. (2007). Approximately normal tests for equal predictive accuracy in nested models. *Journal of Econometrics*, 138(1), 291–311.
- ▣ Corsi, F. (2009). A simple approximate long-memory model of realized volatility. *Journal of Financial Econometrics*, 7(2), 174–196.
- ▣ Das, A., Kong, W., Sen, R., Zhou, Y. (2024). A decoder-only foundation model for time-series forecasting. *Proceedings of ICML*, PMLR 235; arXiv:2310.10688.
- ▣ Diebold, F.X. (2015). Comparing predictive accuracy, twenty years later: a personal perspective on the use and abuse of Diebold–Mariano tests. *Journal of Business & Economic Statistics*, 33(1), 1.
- ▣ Diebold, F.X., Gunther, T.A., Tay, A.S. (1998). Evaluating density forecasts with applications to financial risk management. *International Economic Review*, 39(4), 863–883.
- ▣ Diebold, F.X., Mariano, R.S. (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3), 253–263.
- ▣ Engle, R.F., Manganelli, S. (2004). CAViaR: conditional autoregressive value at risk by regression quantiles. *Journal of Business & Economic Statistics*, 22(4), 367–381.
- ▣ Fischer, T., Krauss, C. (2018). Deep learning with long short-term memory networks for financial market predictions. *European Journal of Operational Research*, 270(2), 654–669.

References (3/7)

- ▣ Fissler, T., Ziegel, J.F. (2016). Higher order elicibility and Osband's principle. *Annals of Statistics*, 44(4), 1680–1707.
- ▣ Franke, J., Härdle, W.K., Hafner, C.M. (2019). *Statistics of Financial Markets: An Introduction*, 5th ed. Springer.
- ▣ Garza, A., Rosillo, R., Mendoza-Smith, R., Salinas, D., Williams, A.R., Ashok, A. (2026). Impermanent: a live benchmark for temporal generalization in time series forecasting. [arXiv:2603.08707](#).
- ▣ Giacomini, R., White, H. (2006). Tests of conditional predictive ability. *Econometrica*, 74(6), 1545–1578.
- ▣ Gibbs, I., Candès, E. (2021). Adaptive conformal inference under distribution shift. *NeurIPS*; [arXiv:2106.00170](#).
- ▣ Gneiting, T. (2011). Making and evaluating point forecasts. *Journal of the American Statistical Association*, 106(494), 746–762.
- ▣ Gneiting, T., Raftery, A.E. (2007). Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477), 359–378.
- ▣ Goel, A., Pasricha, P., Magris, M., Kannianen, J. (2025). Foundation time-series AI model for realized volatility forecasting. [arXiv:2505.11163](#).
- ▣ Gu, S., Kelly, B., Xiu, D. (2020). Empirical asset pricing via machine learning. *Review of Financial Studies*, 33(5), 2223–2273.
- ▣ Hansen, P.R. (2005). A test for superior predictive ability. *Journal of Business & Economic Statistics*, 23(4), 365–380.

References (4/7)

- ▣ Hansen, P.R., Lunde, A., Nason, J.M. (2011). [The model confidence set](#). *Econometrica*, 79(2), 453–497.
- ▣ Härdle, W.K., Chen, C.Y.-H., Overbeck, L. (eds.) (2017). [Applied Quantitative Finance](#), 3rd ed. Springer.
- ▣ Hochreiter, S., Schmidhuber, J. (1997). [Long short-term memory](#). *Neural Computation*, 9(8), 1735–1780.
- ▣ Hornik, K., Stinchcombe, M., White, H. (1989). [Multilayer feedforward networks are universal approximators](#). *Neural Networks*, 2(5), 359–366.
- ▣ Kelly, B., Malamud, S., Zhou, K. (2024). [The virtue of complexity in return prediction](#). *Journal of Finance*, 79(1), 459–503.
- ▣ Kingma, D.P., Ba, J. (2015). [Adam: a method for stochastic optimization](#). *ICLR*; arXiv:1412.6980.
- ▣ LeCun, Y., Bengio, Y., Hinton, G. (2015). [Deep learning](#). *Nature*, 521(7553), 436–444.
- ▣ Liu, C., Aksu, T., Liu, J., Liu, X., et al. (2025). [Moirai 2.0: when less is more for time series forecasting](#). arXiv:2511.11698.
- ▣ Lo, A.W., MacKinlay, A.C. (1990). [An econometric analysis of nonsynchronous trading](#). *Journal of Econometrics*, 45(1–2), 181–211.
- ▣ Lopez-Lira, A., Tang, Y., Zhu, M. (2025). [The memorization problem: can we trust LLMs' economic forecasts?](#) arXiv:2504.14765.

References (5/7)

- ▣ McNeil, A.J., Frey, R. (2000). [Estimation of tail-related risk measures for heteroscedastic financial time series: an extreme value approach](#). *Journal of Empirical Finance*, 7(3–4), 271–300.
- ▣ Meyer, M., Kaltenpoth, S., Albers, H., Zalipski, K., Müller, O. (2025). [TS-Arena: a live forecast pre-registration platform](#). [arXiv:2512.20761](#).
- ▣ Nie, Y., Nguyen, N.H., Sinthong, P., Kalagnanam, J. (2023). [A time series is worth 64 words: long-term forecasting with Transformers](#). *ICLR*; [arXiv:2211.14730](#).
- ▣ Nolde, N., Ziegel, J.F. (2017). [Elicitability and backtesting: perspectives for banking regulation](#). *Annals of Applied Statistics*, 11(4), 1833–1874.
- ▣ Oreshkin, B.N., Carpov, D., Chapados, N., Bengio, Y. (2020). [N-BEATS: neural basis expansion analysis for interpretable time series forecasting](#). *ICLR*; [arXiv:1905.10437](#).
- ▣ Pan, Z., Ezzat, A.A. (2026). [Evaluating time series foundation models for electricity price forecasting: contamination risk, distributional shifts, and covariate dependence](#). [arXiv:2607.02623](#).
- ▣ Pascanu, R., Mikolov, T., Bengio, Y. (2013). [On the difficulty of training recurrent neural networks](#). *ICML*; [arXiv:1211.5063](#).
- ▣ Patton, A.J. (2011). [Volatility forecast comparison using imperfect volatility proxies](#). *Journal of Econometrics*, 160(1), 246–256.
- ▣ Patton, A.J., Ziegel, J.F., Chen, R. (2019). [Dynamic semiparametric models for expected shortfall \(and value-at-risk\)](#). *Journal of Econometrics*, 211(2), 388–413.
- ▣ Pele, D.T., Mazurencu-Marinescu-Pele, M. (2026). [Finite-sample precision limits for expected shortfall forecast comparisons](#). *Mathematics*, 14(13), 2316.

References (6/7)

- ▣ Pele, D.T., Wesselhöfft, N., Härdle, W.K., Kolossiatis, M., Yatracos, Y.G. (2023). *Are cryptos becoming alternative assets?* *The European Journal of Finance*, 29(10), 1064–1105.
- ▣ Rahimi, A., Recht, B. (2007). *Random features for large-scale kernel machines.* *Advances in Neural Information Processing Systems*, 20.
- ▣ Rahimikia, E., Ni, H., Wang, W. (2025). *Re(Visiting) time series foundation models in finance.* arXiv:2511.18578.
- ▣ Rasul, K., Ashok, A., Williams, A.R., Ghonia, H., et al. (2023). *Lag-Llama: towards foundation models for probabilistic time series forecasting.* arXiv:2310.08278.
- ▣ Romano, J.P., Wolf, M. (2005). *Stepwise multiple testing as formalized data snooping.* *Econometrica*, 73(4), 1237–1282.
- ▣ Rumelhart, D.E., Hinton, G.E., Williams, R.J. (1986). *Learning representations by back-propagating errors.* *Nature*, 323(6088), 533–536.
- ▣ Salinas, D., Flunkert, V., Gasthaus, J., Januschowski, T. (2020). *DeepAR: probabilistic forecasting with autoregressive recurrent networks.* *International Journal of Forecasting*, 36(3), 1181–1191.
- ▣ Tan, M., Merrill, M.A., Gupta, V., Althoff, T., Hartvigsen, T. (2024). *Are language models actually useful for time series forecasting?* *NeurIPS*; arXiv:2406.16964.
- ▣ Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., et al. (2017). *Attention is all you need.* *NeurIPS*; arXiv:1706.03762.
- ▣ Welch, I., Goyal, A. (2008). *A comprehensive look at the empirical performance of equity premium prediction.* *Review of Financial Studies*, 21(4), 1455–1508.

References (7/7)

- White, H. (2000). [A reality check for data snooping](#). *Econometrica*, 68(5), 1097–1126.
- Woo, G., Liu, C., Kumar, A., Xiong, C., Savarese, S., Sahoo, D. (2024). [Unified training of universal time series forecasting Transformers](#). *ICML*; arXiv:2402.02592.